

Loading the Lookup Table

Steps to load the relevant data in the Lookup Table

1. Create an external table **card_member** by using hive query language and store in in hadoop location.
2. Create an external table **member_score** by using hive query language and store in in hadoop location.
3. Load the data into hive tables from hadoop file system which contain extracted data from RDS DB.
4. Insert the data into **look_up_table** table which was created in Hbase system from **card_member** and **member_score** table after applying join on tables to get Moving average and Standard deviation using hql.
5. Best way is to create **preAnalysis.hql** file with below queries and run it by using **hive -f preAnalysis.hql** command.

```
CREATE EXTERNAL TABLE IF NOT EXISTS card_member (  
  card_id STRING,  
  member_id STRING,  
  member_joining_dt STRING,  
  card_purchase_dt STRING,  
  country STRING,  
  city STRING  
)  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY ','  
LOCATION '/user/hadoop/RDS/card_member';  
  
CREATE EXTERNAL TABLE IF NOT EXISTS member_score (  
  member_id STRING,  
  score INT  
)  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY ','  
LOCATION '/user/hadoop/RDS/member_score';  
  
LOAD DATA INPATH  
'hdfs:/user/hadoop/sqoop/import/cred_financials_data/card_member'  
OVERWRITE INTO TABLE card_member;
```

```
LOAD DATA INPATH
'hdfs:/user/hadoop/sqoop/import/cred_financials_data/member_score'
OVERWRITE INTO TABLE member_score;
```

- Creating lookup table with Moving average and standard deviation of the last 10 transactions for each card_id.

```
Insert into look_up_table select ltt.card_id,
cm.card_purchase_dt, ltt.transaction_dt, ltt.member_id,
member_joining_dt, country, city, UCL, ltt.postcode, score
from last_ten_transactions ltt
inner join
member_score ms
on ltt.member_id=ms.member_id and ltt.ranking=1
inner join
card_member cm
on cm.member_id=ltt.member_id and ltt.ranking=1
inner join
(select card_id, avg(amount)+ (3* stddev(amount)) as UCL
from last_ten_transactions
where ranking<=10
group by card_id
) as ucl
on ltt.card_id=ucl.card_id ;

exit;
```

Command to see the table created and the content.

Hive:

```
show tables
cc
```

Hbase:

```
List
count 'look_up_table'
scan 'look_up_table'
```

Screenshot of the created table

- Execution & Output of preAnalysis.hql file execution with all above queries:

```
[hadoop@ip-172-31-5-80 ~]$ vi preAnalysis.hql
[hadoop@ip-172-31-5-80 ~]$ cat preAnalysis.hql
CREATE EXTERNAL TABLE IF NOT EXISTS card_member (
  card_id STRING,
  member_id STRING,
  member_joining_dt STRING,
  card_purchase_dt STRING,
  country STRING,
  city STRING
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
LOCATION '/user/hadoop/RDS/card_member';

CREATE EXTERNAL TABLE IF NOT EXISTS member_score (
  member_id STRING,
  score INT
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
LOCATION '/user/hadoop/RDS/member_score';

LOAD DATA INPATH 'hdfs://user/hadoop/sqoop/import/cred_financials_data/card_member' OVERWRITE INTO TABLE card_member;
LOAD DATA INPATH 'hdfs://user/hadoop/sqoop/import/cred_financials_data/member_score' OVERWRITE INTO TABLE member_score;

Insert into look_up_table select ltt.card_id, cm.card_purchase_dt, ltt.transaction_dt, ltt.member_id, member_joining_dt, country, city, UCL, ltt.postcode, score
from last_ten_transactions ltt
inner join
member_score ms
on ltt.member_id=ms.member_id and ltt.ranking=1
inner join
card_member cm
on cm.member_id=ltt.member_id and ltt.ranking=1
inner join
(select card_id, avg(amount)+ (3* stddev(amount)) as UCL
from last_ten_transactions
where ranking<=10
group by card_id
) as ucl
on ltt.card_id=ucl.card_id ;

exit;
```

```
[hadoop@ip-172-31-5-80 ~]$ hive -f preAnalysis.hql

Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j2.properties Async: true
OK
Time taken: 0.912 seconds
OK
Time taken: 0.038 seconds
Loading data to table default.card_member
OK
Time taken: 0.702 seconds
Loading data to table default.member_score
OK
Time taken: 0.367 seconds
No Stats for default@card_transactions, Columns: member_id, amount, postcode, transaction_dt, card_id, status
No Stats for default@member_score, Columns: member_id, score
No Stats for default@card_member, Columns: member_id, country, member_joining_dt, city, card_purchase_dt
No Stats for default@card_transactions, Columns: amount, transaction_dt, card_id, status
Query ID = hadoop_20240119142540_7f492c8c-a6fb-4889-a504-acc882380a9a
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1705670818466_0010)

-----
      VERTICES      MODE      STATUS      TOTAL      COMPLETED      RUNNING      PENDING      FAILED      KILLED
-----
Map 1 ..... container      SUCCEEDED      1          1          0          0          0          0
Map 3 ..... container      SUCCEEDED      1          1          0          0          0          0
Map 4 ..... container      SUCCEEDED      1          1          0          0          0          0
Map 5 ..... container      SUCCEEDED      1          1          0          0          0          0
Reducer 2 ..... container      SUCCEEDED      2          2          0          0          0          0
Reducer 6 ..... container      SUCCEEDED      2          2          0          0          0          0
Reducer 7 ..... container      SUCCEEDED      2          2          0          0          0          0
-----
VERTICES: 07/07 [=====>>] 100% ELAPSED TIME: 12.10 s
-----
OK
Time taken: 20.524 seconds
Hadoop@ip-172-31-5-80 ~$
```

Table data verification

- Hive: Table has been created with 999 rows.

```

Time taken: 27.072 seconds
[hadoop@ip-172-31-4-180 ~]$ hive

Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j2.properties Async: true
hive> show tables
> ;
OK
card_member
card_transactions
card_transactions_stg
last_ten_transactions
look_up_table
member_score
Time taken: 0.524 seconds, Fetched: 6 row(s)
hive> select count(*) from card_member;
Query ID = hadoop_20240118100958_1f994c0a-4b92-47a5-9040-ff662e714b92
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1705566423041_0027)

-----
VERTICES      MODE           STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
-----
Map 1 ..... container  SUCCEEDED    1         1         0         0         0         0
Reducer 2 ..... container  SUCCEEDED    1         1         0         0         0         0
-----
VERTICES: 02/02 [=====>>>] 100% ELAPSED TIME: 4.58 s
-----
OK
999
Time taken: 7.456 seconds, Fetched: 1 row(s)
hive> select count(*) from member_score;
Query ID = hadoop_20240118101018_25f90bfa-a5aa-441d-94fb-70a7b025301d
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1705566423041_0027)

-----
VERTICES      MODE           STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
-----
Map 1 ..... container  SUCCEEDED    1         1         0         0         0         0
Reducer 2 ..... container  SUCCEEDED    1         1         0         0         0         0
-----
VERTICES: 02/02 [=====>>>] 100% ELAPSED TIME: 4.50 s
-----
OK
999
Time taken: 5.294 seconds, Fetched: 1 row(s)

```

Hbase:

- Table has been created with 999 rows in Hbase system.

```
[hadoop@ip-172-31-4-180 ~]$ sudo hbase shell
HBase Shell
Use "help" to get list of supported commands.
Use "exit" to quit this interactive shell.
Version 1.4.13, rUnknown, Fri Apr 17 15:18:24 UTC 2020

hbase(main):001:0> list
TABLE
card_transactions
look_up_table
2 row(s) in 0.2080 seconds

=> ["card_transactions", "look up table"]
hbase(main):002:0> count 'look up table'
999 row(s) in 0.2170 seconds

=> 999
hbase(main):003:0> █
```

```
6595814135833988 column=Rule_details:member_joining_dt, timestamp=1705572542342, value=2012-02-10 00:15:10.0
6595814135833988 column=Rule_params:UCL, timestamp=1705572542342, value=1.3626273240525039E7
6595814135833988 column=Rule_params:postcode, timestamp=1705572542342, value=22508
6595814135833988 column=Rule_params:score, timestamp=1705572542342, value=210
6595814135833988 column=card_details:card_purchase_dt, timestamp=1705572542342, value=06/14
6595814135833988 column=card_details:transaction_dt, timestamp=1705572542342, value=30-01-2018 02:03:54
6595928469079750 column=Location:city, timestamp=1705572542342, value=Raleigh
6595928469079750 column=Location:country, timestamp=1705572542342, value=United States
6595928469079750 column=Member_details:member_id, timestamp=1705572542342, value=422173403853722
6595928469079750 column=Member_details:member_joining_dt, timestamp=1705572542342, value=2012-02-10 02:19:20.0
6595928469079750 column=Rule_params:UCL, timestamp=1705572542342, value=1.142797041440079E7
6595928469079750 column=Rule_params:postcode, timestamp=1705572542342, value=98349
6595928469079750 column=Rule_params:score, timestamp=1705572542342, value=412
6595928469079750 column=card_details:card_purchase_dt, timestamp=1705572542342, value=02/14
6595928469079750 column=card_details:transaction_dt, timestamp=1705572542342, value=24-01-2018 12:38:22
6597703848279563 column=Location:city, timestamp=1705572542342, value=Pinewood
6597703848279563 column=Location:country, timestamp=1705572542342, value=United States
6597703848279563 column=Member_details:member_id, timestamp=1705572542342, value=657991752750272
6597703848279563 column=Member_details:member_joining_dt, timestamp=1705572542342, value=2017-02-16 07:45:34.0
6597703848279563 column=Rule_params:UCL, timestamp=1705572542342, value=1.4918634149498457E7
6597703848279563 column=Rule_params:postcode, timestamp=1705572542342, value=95699
6597703848279563 column=Rule_params:score, timestamp=1705572542342, value=218
6597703848279563 column=card_details:card_purchase_dt, timestamp=1705572542342, value=03/17
6597703848279563 column=card_details:transaction_dt, timestamp=1705572542342, value=27-01-2018 10:51:49
6598830758632447 column=Location:city, timestamp=1705572542342, value=Tampa
6598830758632447 column=Location:country, timestamp=1705572542342, value=United States
6598830758632447 column=Member_details:member_id, timestamp=1705572542342, value=660350842993890
6598830758632447 column=Member_details:member_joining_dt, timestamp=1705572542342, value=2013-07-26 02:22:17.0
6598830758632447 column=Rule_params:UCL, timestamp=1705572542342, value=1.2227949982601807E7
6598830758632447 column=Rule_params:postcode, timestamp=1705572542342, value=19421
6598830758632447 column=Rule_params:score, timestamp=1705572542342, value=293
6598830758632447 column=card_details:card_purchase_dt, timestamp=1705572542342, value=12/16
6598830758632447 column=card_details:transaction_dt, timestamp=1705572542342, value=30-01-2018 00:18:34
6599900931314251 column=Location:city, timestamp=1705572542342, value=Loves Park
6599900931314251 column=Location:country, timestamp=1705572542342, value=United States
6599900931314251 column=Member_details:member_id, timestamp=1705572542342, value=928036864799687
6599900931314251 column=Member_details:member_joining_dt, timestamp=1705572542342, value=2017-09-09 03:31:54.0
6599900931314251 column=Rule_params:UCL, timestamp=1705572542342, value=1.2121408572464656E7
6599900931314251 column=Rule_params:postcode, timestamp=1705572542342, value=97423
6599900931314251 column=Rule_params:score, timestamp=1705572542342, value=267
6599900931314251 column=card_details:card_purchase_dt, timestamp=1705572542342, value=10/17
6599900931314251 column=card_details:transaction_dt, timestamp=1705572542342, value=31-01-2018 11:25:16

99 row(s) in 9.1220 seconds
hbase(main):004:0> █
```